

Cryptanalysis Report: The Vigenère Cipher

Introduction: The Vulnerability of Periodicity

The Vigenère cipher was long considered "indecipherable" because it successfully obscures the letter frequencies of the original message, rendering simple frequency analysis useless. However, its security is entirely dependent on its key. The cipher's fundamental weakness is the **periodic nature of its keyword**. Because the keyword is shorter than the message and is repeated, it creates subtle but detectable statistical patterns in the ciphertext. A Vigenère cipher using a non-repeating key as long as the message itself would be a One-Time Pad, which is provably unbreakable. The Vigenère is breakable precisely because its key is not a one-time pad.

The cryptanalytic attack is a two-phase process:

1. Determine the Keyword Length.
2. Determine the Keyword Itself.

Phase I: Determining the Keyword Length

Method A: Kasiski Examination

The Kasiski Examination identifies the keyword length by finding repeated sequences within the ciphertext. If the same plaintext sequence is encrypted by the same sequence of keyword letters, it produces an identical ciphertext sequence. The distance between these repetitions is usually a multiple of the keyword length (L).

Steps:

1. Scan the ciphertext for identical sequences of three or more characters.
2. Measure the distances between the start of each repeated sequence.
3. Factor all distances found. The keyword length is likely the greatest common divisor (GCD) of these distances.

Example:

If repeated sequences have distances of 30 (factors: 1,2,3,5,6,10,15,30) and 42 (factors: 1,2,3,6,7,14,21,42), the GCD is 6 — likely the key length.

Method B: Index of Coincidence (IC)

The Index of Coincidence (IC) measures how uneven a letter frequency distribution is. Plain English has an $IC \approx 0.067$; random text ≈ 0.0385 . If we guess the correct key length L and arrange the ciphertext into L columns, each column will be a Caesar cipher of a subset of the plaintext.

Steps:

1. Assume a key length L (start from 1,2,3,...).
2. Split the ciphertext into L columns.
3. Calculate the IC for each column using: $IC = \frac{\sum [n_i(n_i-1)]}{[N(N-1)]}$

4. Average the ICs. The key length giving an average closest to 0.067 is the correct one.

Phase II: Keyword Recovery

Once L is known, treat the ciphertext as L interleaved Caesar ciphers. Each column's shift key can be found using the Chi-Squared (χ^2) statistic.

Chi-Squared (χ^2) Statistic Method

For each column, try all 26 possible shifts (A–Z). For each shift, decrypt and compute:

$$\chi^2 = \sum [(O_i - E_i)^2 / E_i]$$

where O_i = observed count of letter i, and E_i = expected count in English.

The shift with the lowest χ^2 is the correct one. Repeat for all columns to recover the keyword.

Once the keyword is found, decrypting the ciphertext is straightforward.